Multimodal IMDB Genre Classification with Keras Report

1 Introduction

This project investigates a multimodal method for classifying film genre using textual and visual data. Each film may belong to multiple genres simultaneously; thus, the work is formulated as a multilabel classification problem. Two deep learning models are created: a **Bidirectional Long Short-Term Memory** network for genre classification from textual overviews and a **Convolutional Neural Network** for genre classification from film posters.

The objective is to compare the performance of each modality and evaluate how effectively it contributes to genre prediction. To provide a comprehensive evaluation, the models are assessed using qualitative and quantitative analysis.

2 Methodology

2.1 Dataset and Preprocessing

The dataset is made up of movie posters, textual overviews, and multi-label genre annotations. Poster images are converted to *float32* format and resized to *64 by 64* pixels. To enhance generalisation, data augmentation techniques such as random flipping and brightness and contrast modifications were utilised. Batching, caching, and prefetching are also used to optimise the dataset. Also, a TextVectorization layer with a vocabulary size of 10,000 is utilised in processing textual overviews. In order to identify relevant vocabulary patterns, the encoder is adapted on the training data.

2.2 CNN

The CNN model extracts visual information using residual blocks, which enhance training stability and enable deeper representations. Two convolutional layers with skip connections and dropout are applied in each block.

Pooling decreases spatial dimensions while increasing feature depth from 32 to 128 filters. A dropout and a dense layer are used after global average pooling to reduce overfitting, and multi-label predictions are produced by a sigmoid output layer.

2.3 LSTM Model for Overview Classification

An embedding layer followed by two stacked bidirectional LSTM layers is used by the LSTM model to process textual data. As a result, contextual information can be captured by the model in both directions.

The final dense layer uses sigmoid activation to produce genre probabilities, while dropout is used to reduce overfitting.

3 Model Performance and Evaluation

3.1 CNN Model Performance

Throughout training, the CNN model demonstrates steady and consistent learning behaviour. As shown in **Figure 1**, training precision rises steadily across epochs, while validation precision remains comparatively constant and slightly lower. This suggests that the model is learning meaningful visual patterns without significant overfitting.

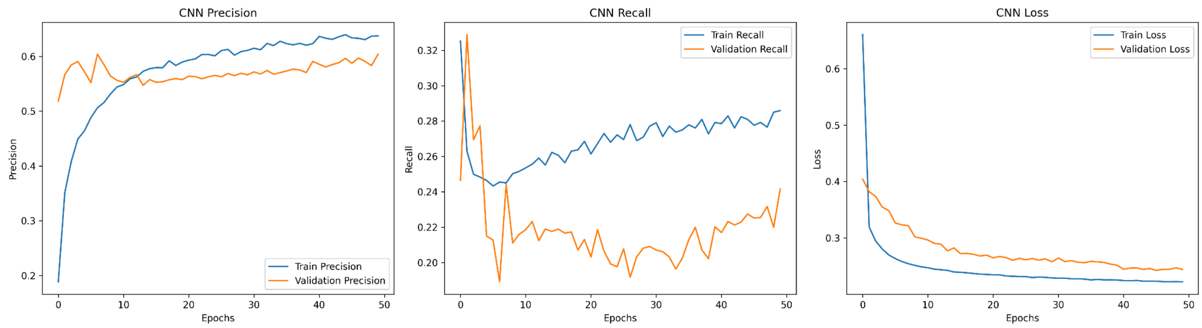

Figure 1: CNN Precision, Recall, and Loss Plots

This observation is further supported by the loss curves, which show that there is only a slight difference between the training and validation losses with time. This suggests that the model does a respectable job of generalising to unseen data.

However, on the validation set in particular, recall is noticeably lower than precision. This indicates that the model is conservative in its predictions: it frequently predicts a genre correctly, but it fails to identify many of the true labels. This behaviour draws attention to a limitation in the context of multi-label classification since the model fails to adequately capture all relevant genres for each film.

Overall, although the CNN model achieves reasonable generalisation and consistent convergence, its lower recall indicates that it struggles to recover the entire set of true labels.

3.2 LSTM Model Performance

In contrast to CNN, the LSTM model shows a distinct learning pattern. As illustrated in **Figure 2**, training accuracy rapidly rises, and training loss constantly falls, indicating that the model is successfully learning patterns from the textual data.

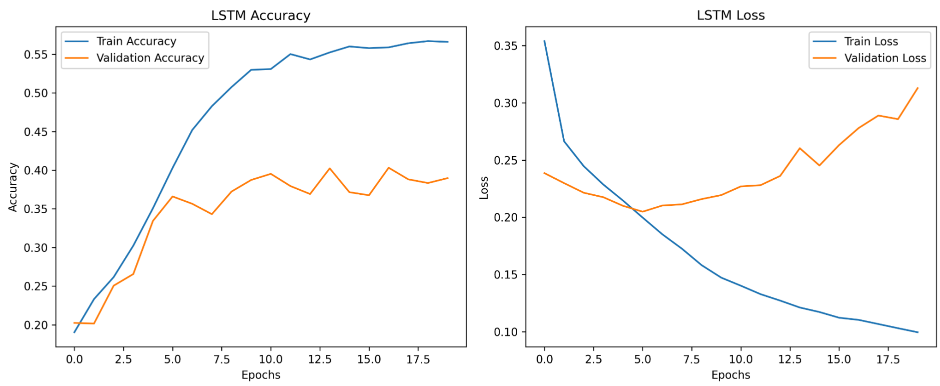

Figure 2: LSTM Accuracy and Loss Plots

Validation performance is visibly less stable, as it is seen that as validation loss starts to rise as training goes on, validation accuracy initially improves but finally plateaus. The presence of overfitting, in which the model grows more specialised to the training data without enhancing its performance on unseen samples, is suggested by this gap between training and validation measurements.

The LSTM model appears to be less consistent in generalisation and more sensitive to overfitting than CNN. Although it is capable of extracting semantic information from the overviews, improved predictive performance on the validation set is not usually the result.

3.3 Qualitative and Comparative Analysis

By comparing the top three predicted genres from each model with the ground truth labels, a set of example films was examined in order to obtain a deeper insight into the model's behaviour.

	Film Index	Ground Truth	CNN Top 3	LSTM Top 3	CNN Hits	LSTM Hits	Comment
0	0	Comedy, Drama, Romance	Drama, Romance, Comedy	Drama, Comedy, Romance	3	3	Similar performance
1	10	Adventure, Comedy, Family	Drama, Comedy, Romance	Drama, Action, Comedy	1	1	Similar performance
2	25	Comedy, Drama, Sport	Comedy, Drama, Romance	Drama, Comedy, Romance	2	2	Similar performance
3	50	Comedy	Drama, Comedy, Romance	Drama, Comedy, Romance	1	1	Similar performance
4	100	Mystery, Thriller	Drama, Horror, Thriller	Drama, Crime, Comedy	1	0	CNN stronger

Table 1: Example Summary Table

Film Example 0


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FILM INDEX: 0
GROUND TRUTH GENRES: Comedy, Drama, Romance

OVERVIEW:
Doug's a concierge at a luxury hotel on Manhattan. He saves all his tips towards his plan for a hotel. A
potential investor seduces the girl, Doug loves, with false promises of leaving his wife. Doug's dilemma:
hotel project or girl?

CNN TOP 3 PREDICTIONS:
  - Drama (0.647) yes
  - Romance (0.356) yes
  - Comedy (0.342) yes

LSTM TOP 3 PREDICTIONS:
  - Drama (0.647) yes
  - Comedy (0.632) yes
  - Romance (0.555) yes

CNN top-3 hits: 3/3
LSTM top-3 hits: 3/3
==========================================================================================
```

Example Outputs (Posters + Predictions)

Both models demonstrate broadly comparable performance throughout the chosen examples. In most cases, each model typically accurately recognises at least one genre within its top three predictions, and in two instances, both models correctly identify two genres and three genres correctly. This suggests that partial information about the underlying genre distribution can be captured by both models.

Figure 3, which shows the number of correct genres within the top three predictions for each example, provides a clearer comparison.

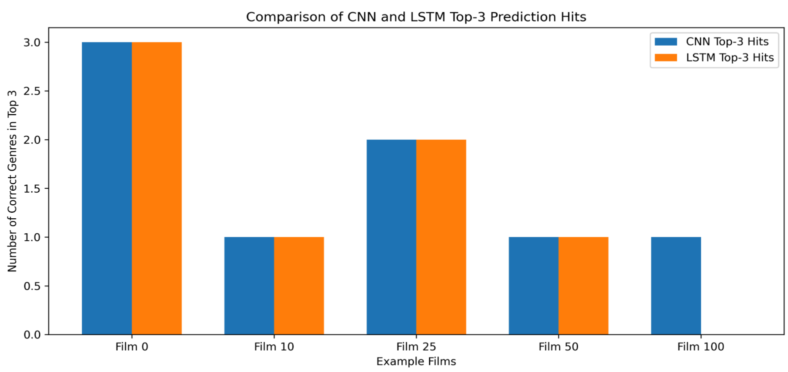

Figure 3: CNN vs LSTM Top-3 Hits

The findings show that neither model consistently outperforms the other in most instances. In one instance, the CNN shows a slight advantage, suggesting that visual features may offer more distinct signals for certain genres. However, overall performance remains extremely comparable, with both models achieving only partial correctness. This demonstrates the difficulty of multi-label classification, where it is common to predict a subset of correct labels but challenging to capture the entire set of genres.

3.4 Genre Distribution and Prediction Behaviour

The test set's genre distribution is highly imbalanced, with some genres, like comedy and drama, appearing much more frequently than others.

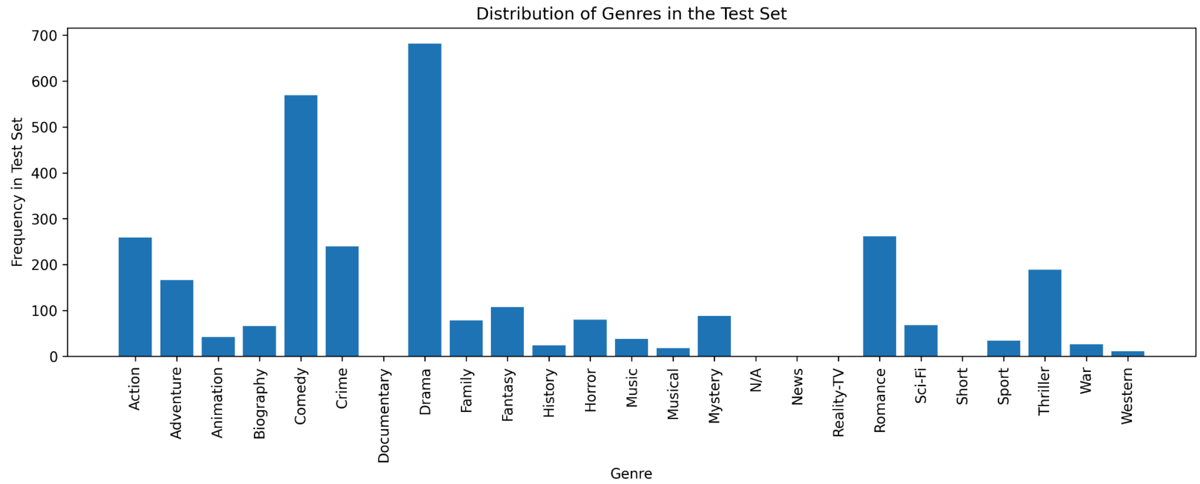

Figure 4: Genre Distribution Plot

This imbalance, as observed in **Figure 4**, affects model behaviour, as both models are more likely to learn and predict popular genres while having difficulties with less frequent ones.

Figure 5 shows a comparison of the expected and ground truth genre frequencies.

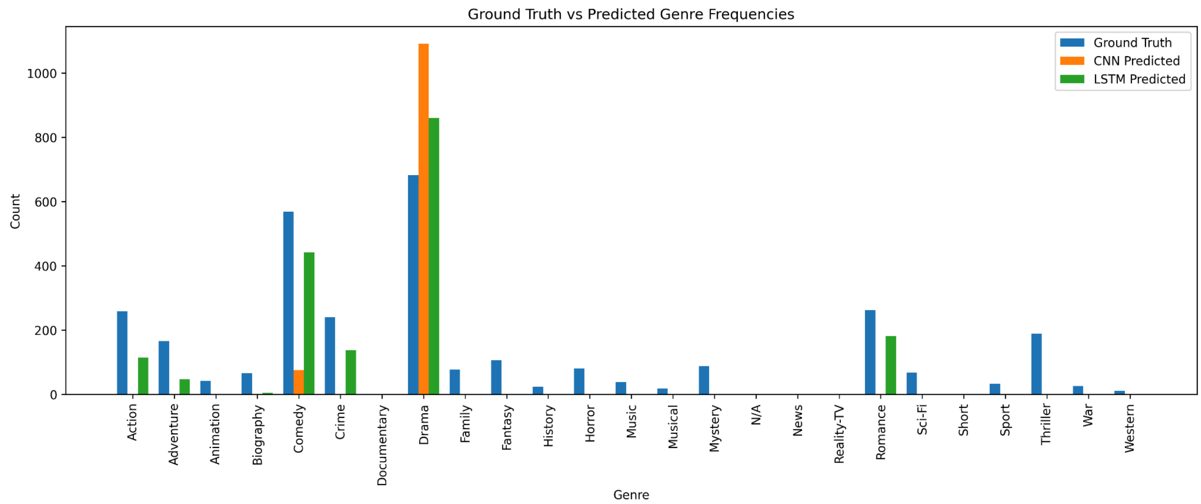

Figure 5: Ground Truth vs Predicted Frequencies

Both models show a significant bias toward popular genres, particularly the '*Drama'* genre. Relative to its actual frequency, this genre appears to be overpredicted by CNN most especially, whereas many other genres are either underpredicted or not predicted at all. Though slightly less dramatic, the LSTM exhibits a similar trend.

This behaviour reflects the effects of class imbalance and indicates that both models are more adept at learning common genre patterns than identifying rarer categories.

4. Critical Evaluation

The results emphasise the challenges associated with multi-label genre classification, in which one film may belong to numerous categories (more than one). This requires models to capture a broader and more nuanced set of features, making it challenging to achieve perfect prediction accuracy.

Class imbalance is the major factor influencing performance. The dataset is overwhelmingly dominated by genres like comedy and drama, although other genres are less common. Both models exhibit a bias towards these dominating classes, frequently over-predicting them while under-representing less common genres. Their ability to generalise across all categories is therefore gravely limited.

CNN demonstrates better generalisation and more stable training. While its lower recall indicates that many relevant labels are overlooked, its higher precision indicates accurate predictions when a genre is identified.

The LSTM, on the other hand, suffers from overfitting despite learning powerful patterns from the training data. Weaker generalisation is indicated by the gap between training and validation performance. Despite the fact that textual data offers rich semantic information, genre indicators may not always be clear.

Overall, both models attain partial correctness and accuracy, usually identifying some but not all relevant genres, suggesting that each partially captures only part of the underlying data.

5. Conclusion

This project shows how deep learning methods may be applied to multimodal genre categorisation using both textual and visual data. The CNN model, which was trained on movie posters, demonstrates improved generalisation and steady learning, especially when it comes to recognising visually diverse genres. Although it is more likely to overfit, the LSTM model, which was trained on textual overviews, can capture semantic information.

Overall, both models achieve average performance and are able to identify relevant genres to some extent, but neither reliably captures the full set of labels. The outcomes highlight the challenge of class imbalance and the difficulty of multi-label classification.

Future improvements could incorporate combining both modalities into a single hybrid model and using techniques (such as weighted loss functions or data resampling) to address class imbalance. These methods may lead to more accurate and balanced genre predictions that are more impartial and do not skew sensitively to the data’s imbalance.